Software

Application Software

help user to solve particular problems

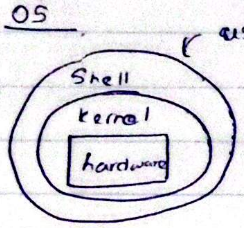
System Software

collection of programs, which helps users run other programs

- Handling user requests
- Managing I/O
- File managment
- Program translation from source

- OS (windows, LINUX, MAC)
- Compilers and assemblers
- Linkers loaders
- Editors, debugger.

OS



interface between computer hardware ~~bus~~ and users

tasks

Assigning memory and disk space to program and data files

Moving data between I/O devices, memory, disk units

Handling I/O operations, with parallel operation

Handling multiple user interactions

Computer Architecture

Von - Neuman Architecture

- Instructions and data are stored in the same memory module
- Suitable for most general purpose processors

Disadvantages

The processor - memory bus acts as the bottleneck

All instructions and data are moved back and forth through the pipe

What is bottleneck

A hardware performance limit where extremly fast CPU is forced to slow down because it has to wait data to travel from memory.

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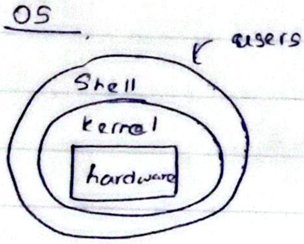

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The cause

Program instructions and data share the exact same physical wire pathway (bus). The CPU cannot read both at same time and must wait for data to cross the single line

The problem

CPU have gotten vastly faster for decades but RAM haven't kept pace, creating massive speed gap.

Solutions

- Using modified harvard design

Harvard Architecture

Instruction – Program memory

data – data memory

Instructions and data accesses parallely.

bottleneck still remains

Emerging Architectures

Memristors can be used in high capacity, ~~non - voltal~~ non volatile resistive memory systems

Memristors within the memory can also be controlled to carry out some computations

Non volatile Resistive memory

Computation within memory

Memristors (memory resistor)

It is revolutionary electrical component that changes resistance based on current flowed

Simply :- resistor that remember its past

Pipeline in Executing Instructions

Pipeline is an instruction execution technique where multiple program instructions are overlapped in execution, breaking down tasks into sequence of smaller stages so that different parts of the CPU can process different instructions ~~symat~~ simultaneously.

1. Instruction Fetch (IF) – Fetch instruction from memory
2. Instruction Decode (ID) – Decode instruction and reading registers
3. ALU Operation (EX) – Performing arithmetic / logical Operations
4. Memory Access (MEM) – Reading data from or writing to main memory
5. Write Back (WB) – Writing the final result back

Memoristors

1. Processing inside Memory.

   Can store a value and modify electrical signal passing through it. Can process data where it live. no need to send bits back and forth across slow bus.

2. Non volatile – fast

   retains data when power off, operate speed close to main RAM.

3. Analog states – Unlike other transistors (1, 0), this can set to any intermediate resistance value along a continuous scale.